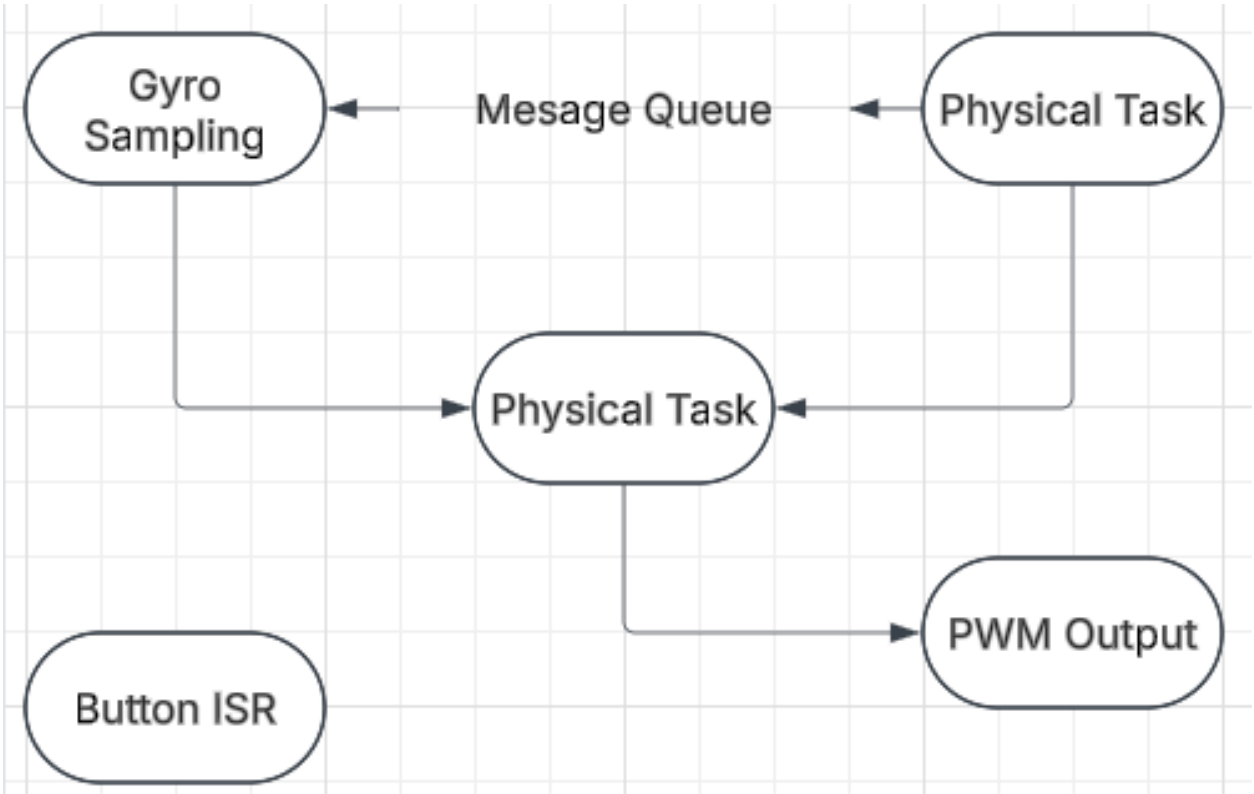
Joel J. Ramos Silva
ECEN 3753

Week 3

Step 1: Task Diagram



For the task diagram the gyro sampling Button ISR have been implemented and this week I will be working on the physical tasks.

Step 2: Project Progress and Report

a) Functionality & Usability Summary

This week I implemented the shield and railgun logic with working visual bars. I also began projectile logic with bouncing and height based on energy. The LCD visuals work consistently, and I'm debugging projectile boundaries and cleanup. I can now press and hold the button to charge the railgun, fire a beam, or double-tap to activate a visual shield that drains capacitor energy over time.

b) Summary of Effort & Estimates

I have completed 60% of my estimated work (21.6 estimated for completed work / 36 hr total estimate) in 77% of the project time (27.8 hr used / 36 hr). For completed tasks, I took 1.29x longer than estimated (27.8 / 21.6).

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Step 3: Task Table

Task Name	Est. Time (hrs)	Actual Time (hrs)	Status
Task Diagram	1.5	1.5	Complete
Button Handling	3	3.5	Complete
Shield Visual & Logic	4	5.0	Complete
Railgun Visual & Logic	4	6.0	Complete
Projectile Logic	5	3.8	In Progress
Display Task Updates	2	2.5	Complete
Cap Energy Bar Logic	2.5	2.5	Complete

Completed this week:

- **Shield Visual & Logic**: I implemented double-tap detection, energy drain every 2 seconds, and shield display using a circle fill. I struggled to get consistent updates, but now it works in real-time.
- **Railgun Visual & Logic**: Beam fires based on hold duration. Adjusted coordinate scaling for energy. Learned how to manage duration-to-distance conversion.
- **Display Update**: Smoothed out display bar behavior. Used vertical stacking of using _ (underscore) chars with dynamic clearing.
- **Projectile Logic (partial)**: Bounce and motion based on energy mostly working. Currently fixing x-position error and ghost trails.

Step 4: Test Plan and Results

a) Unit Tests

- **UT1**: Cut before/after Button_Handler() call. Simulate short and long presses. Tests press-and-hold timing & event generation.
- **UT2**: Cut around Update_Shield() logic. Test if energy drains correctly and shield disables at 0 energy.

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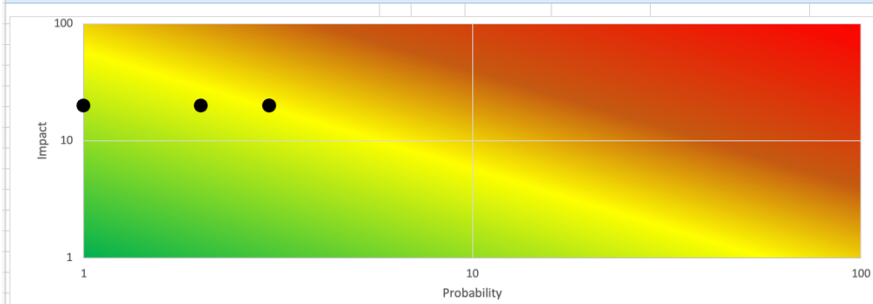
b) Functional Tests

Add/update status ("Pass", "Fail", "NotRun")

Test #	Description	Status
FT1	Button press charges railgun	Pass
FT2	Button release fires beam	Pass
FT3	Beam length based on hold duration	Pass
FT4	Double tap activates shield	Pass
FT5	Shield drains energy every 2s	Pass
FT6	Shield disables at 0 energy	Pass
FT7	Projectile fires with energy-based height	In Progress
FT8	Projectile bounces on cliff and castle	In Progress
FT9	Projectile disappears when off screen	Fail (leaves trail)
FT10	Display shows current cap energy level	Pass

Step 5: Risk Register

Item	P	I	Risk (P*I)	Recognized	Mitigated/ Resolved	ROAM	How
Projectile trails clutter screen	3	20	60	1-Apr-25	Mitigated	M	Clear previous position after movement
Cap energy bar doesn't update fast	2	20	40	2-Apr-25	Resolved	R	Used delay and restructured DisplayTask
Double-tap misfires	3	20	60	3-Apr-25	Mitigated	M	Increased debounce interval
LCD refresh rate too low	1	20	20	4-Apr-25	Resolved	A	Drawing only diffs, not whole screen
			0				
			0				
			0				
			0				
			0